Contents

1 Kinematics

average velocity: $\overline{v} = \frac{\Delta x}{\Delta t}$

instant velocity: $v = lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$

average acceleration: $\overline{a} = \frac{\Delta v}{\Delta t}$

instant velocity: $a = lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$

$x = x_0 + v_0 t$ or $y = y_0 + v_0 t$

$x = x_0 + v_0 t + \frac{1}{2}at^2$ or $y = y_0 + v_0 t + \frac{1}{2}at^2$

position from velocity: $x = x_0 + v(t - t_0)$ and $x = x_0 + t(v_0 + v)$

velocity from acceleration: $v = v_0 + a(t - t_0)$

$x = x_0 + v_0(t - t_0) + \frac{1}{2}a(t - t_0)^2$

$x = \frac{v^2 - v_0^2}{2a}$

$t = \frac{v - v_0}{a}$ projectile motion: when vector of has angle θ with horizontal: $v_x = vcos\theta$ and $v_y = vsin\theta$

Time of flight: $T = \frac{2(v_0 sin\theta_0)}{g}$

Trajectory: $y = (tan\theta_0)x - [\frac{g}{2(v_0 cos\theta_0)^2}]x^2$

Range: $R = \frac{v_0^2 sin2\theta_0}{g}$

1.1 Uniform circular motion

$\theta = \omega t$, where ω is angluar speed, which is constant in this case

$\omega = 2\pi f = \frac{2\pi}{T}$, where f is revolution per second(Hz)

$\overrightarrow{r}(t) = r \cdot cos(\omega)\hat{i} + r \cdot sin(\omega t)\hat{j}$

$\overrightarrow{v}(t) = -\omega r \cdot sin(\omega)\hat{i} + \omega r \cdot cos(\omega t)\hat{j}$

$|\overrightarrow{v}(t)| = \sqrt{(-\omega r \cdot sin(\omega))^2 + (\omega r \cdot cos(\omega t))^2} = \omega r$

$\overrightarrow{a}(t) = -\omega^2 r \cdot cos(\omega)\hat{i} - \omega^2 r \cdot sin(\omega t)\hat{j}$

$|\overrightarrow{a}(t)| = \omega^2 r = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$, where a is centripetal acceleration

Tangential acceleration: $a_T = \frac{d|\overrightarrow{v}|}{dt}$

Total acceleration: $\overrightarrow{a} = \overrightarrow{a_C} + \overrightarrow{a_T}$

$v = \frac{2\pi r}{T}$

1.2 Relative motion

$\overrightarrow{r}_{P/A} = \overrightarrow{r}_{P/B} + \overrightarrow{r}_{B/A}$

$\overrightarrow{v}_{P/A} = \overrightarrow{v}_{P/B} + \overrightarrow{v}_{B/A}$

$v_{P/A-x} = v_{P/B-x} + v_{B/A-x}$

$\overrightarrow{a}_{P/A} = \overrightarrow{a}_{P/B} + \overrightarrow{a}_{B/A}$

$\overrightarrow{v}_{P/A}$ means the velocity of P relative to A.

2 Uncertainties

dependent uncertainties - arise from the same source; independent - arise from different sources

- $A = x + y$:

  $\delta A = \delta x + \delta y$ (dependent)

  $\delta A = \sqrt{\delta x^2 + \delta y^2}$ (independent)

  $\delta x = \sqrt{(\delta x_{random})^2 + (\delta x_s ystematic)^2}$

- $A = x \cdot y$:

  $\frac{\delta A}{|A|} = \frac{\delta x}{|x|} + \frac{\delta y}{|y|}$ (dependent)
  $\frac{\delta A}{|A|} = \sqrt{(\frac{\delta x}{|x|})^2 + (\frac{\delta y}{|y|})^2}$ (independent)

- $A = f(x, y)$:

  $\delta A = |\frac{\partial f}{\partial x}|\delta x + |\frac{\partial f}{\partial y}|\delta y$ (dependent)
  $\delta A = \sqrt{(|\frac{\partial f}{\partial x}|\delta x)^2 + (|\frac{\partial f}{\partial y}|\delta y)^2}$ (independent)

3 Forces

Second Newton's law: $F = ma$

Gravitational force(weight): $F = mg$ $(F = m(g - a))$

Normal force: $N = mg$, for incline plane it would be $N = mgcos\theta$

Kinetic friction: $F_k = \mu_k N$

Static friction: $F_s \leq \mu_s N$

Mass momentum: $\overrightarrow{p} = m\overrightarrow{v}$

Tension: $T = w = mg$

Spring force: $\overrightarrow{F} = -k\overrightarrow{x}$, $F = kx$, where F is restore force

Centripetal force: $F_C = m\frac{v^2}{r} = mr\omega^2$

3.1 less known or needed

Ideal angle of a banked curve: $\theta = tan^{-1}(\frac{v^2}{rg})$

Drag force: $F_D = \frac{1}{2}C\rho Av^2$, where C is drag coefficient and A is area of the object facing the fluid

Terminal velocity: $v_T = \sqrt{\frac{2mg}{\rho CA}}$

Stokes' Law: $F_s = 6\pi r\eta v$, where r is radius of object and η is viscosity of fluid

4 Work and energy

$dW = \overrightarrow{F} \cdot d\overrightarrow{r} = |\overrightarrow{F}||d\overrightarrow{r}|cos\theta$

Work done by force: $W_{AB} = \int_{pathAB} \overrightarrow{F} \cdot d\overrightarrow{r}$

Work done by a varying force or on a curved path: $W = \int_{P_1}^{P_2} \overrightarrow{F} \cdot d\overrightarrow{s} = \int_{P_1}^{P_2} Fcos(\theta)ds = \int_{P_1}^{P_2} F_{||}ds$

Work done by a constant force of kinetic force: $W_{fr} = -f_k \cdot |l_{AB}|$ where $|l_{AB}|$ is path length on the surface

Work done going from A to B by Earth's gavity, near its surface: $W_{grav,AB} = -mg(y_A - y_B)$

Work done from A to B by one-dimensional spring force: $W_{spring,AB} = -\frac{1}{2}k \cdot (x_B^2 - x_A^2)$

Work and energy theorem: $W_{net} = K_B - K_A$

Kinetic energy: $K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$, where p is mass momentum

Gravitational potential energy: $E_pot, g = mgh$

Elastic potential energy: $E_pot, s = \frac{1}{2}kx^2$

Mechanical energy: $E_{mec} = E_{kin} + E_{pot}$

Condition for corservative force in two dimensions: $(\frac{dF_x}{dy}) = (\frac{dF_y}{dx})$

5 Power

rate of doing work: $P = \frac{dW}{dt}$

power as the dot product of force and velocity: $P = \overrightarrow{F} \cdot \overrightarrow{v} = F \cdot v \cdot cos(\theta)$

6 Collisions

Momentum: $\overrightarrow{p} = m\overrightarrow{v}$

Impulse: $d\overrightarrow{J} = \overrightarrow{F}(t)dt$, where F is force applied to an object over some differential time interval

$\overrightarrow{J} = \int_{t_i}^{t_f} \overrightarrow{F}(t)dt \Rightarrow \overrightarrow{J} = \overrightarrow{F}\Delta t \Rightarrow \overrightarrow{J} = m\Delta\overrightarrow{v}$

Impulse - momentum theorem: $\overrightarrow{J} = \Delta\overrightarrow{p}$

$m_1\overrightarrow{v}_{01} + m_2\overrightarrow{v}_{02} = m_1\overrightarrow{v}_1 + m_2\overrightarrow{v}_2$

Inelastic collision: stick together after collision: $0 < K_f < K_i$

Elastic collision: bounce off of each other after collision: $K_f = K_i$

Explosion: $K_f > K_i$

Rocket equation: $v_f = v_i + u \cdot ln(\frac{m_i}{m})$, where u is the exhaust speed(positive), m_i is initial mass of the rocket and m is the mass of the rocket at the time when you want to find its speed

Center of mass: $\overrightarrow{r}_{cm} = \frac{\sum_i m_i\overrightarrow{r}_i}{\sum_i m_i}$

7 Fixed-axis rotation

instantaneous angular velocity: $\omega = \frac{d\theta}{dt}$

Tangential velocity: $v_t = r\omega$, where r is the distance of a particle from the rotation axis

instantaneous angular acceleration: $\alpha = \frac{d^2\theta}{dt^2}$

Tangential acceleration: $a_t = r\alpha$

Radial(centripetal) acceleration and angular speed: $\alpha_{rad} = \frac{v^2}{r} = \omega^2 r$

angular displacement from average angular velocity: $\theta_t = \theta_0 + \overline{\omega}t$

angular velocity from angular acceleration: $\omega_f = \omega_0 + \alpha t$

$\theta_f = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$

$\omega_f^2 = \omega_0^2 + 2\alpha(\Delta\theta)$

$\Delta\theta = \frac{1}{2}(\omega_0 + \omega)t$

Moment of inertia: $I = \sum_i m_i r_i^2$

Rotational kinetic energy: $E_{kin,rot} = \frac{1}{2}I\omega^2$

Parallel-axis theorem: $I_P = I_{cm} + Md^2$, where I_{cm} is the moment of inertia around an axis through the center of mass, I_P is the moment of inertia around an axis parallel to that axis, d is the distance between the two axes, and M is the mass of the body

7.1 Moment of inertia for various shapes

Slender rod, axis through center: $I = \frac{1}{12}ML^2$ where L is length of the rod

Slender rod, axis through one end: $I = \frac{1}{3}ML^2$

Rectangular plate, axis through center: $I = \frac{1}{12}M(a^2 + b^2)$, where a is the width and b is the length of the plate

Rectangular plate, axis along edge: $I = \frac{1}{3}Ma^2$

Hollow cylinder: $I = \frac{1}{2}M(R_1^2 + R_2^2)$, where R_1 is radius of hollow part and R_2 is the radius of the whole cylinder

Solid cylinder: $I = \frac{1}{2}MR^2$

Thin-walled hollow cylinder: $I = MR^2$

Solid sphere: $I = \frac{2}{5}MR^2$

Thin-walled hollow sphere: $I = \frac{2}{3}MR^2$

7.2 Torque

$\overrightarrow{\tau} = \overrightarrow{r} \times \overrightarrow{F}$

$\tau = Fl = rF\sin(\varphi) = F_{\tan}r$

where **l** is the length of the lever arm, **r** is the distance of the point where the force is applied from the axis of rotation, φ is the angle between the force vector and the position vector of the point where the force is applied with respect to the axis of rotation, and $\mathbf{F}_{\tan}$ is the tangential force component.

Newton's second law for rotation: $\sum \tau = I\alpha$, $\overrightarrow{\tau} = \frac{d\overrightarrow{L}}{dt}$

Kinetic energy in combined translation and rotation: $E_{kin} = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

Work done by a torque: $W = \int_{\theta_1}^{\theta_2} \tau_z d\theta$, $W = \tau_z \Delta\theta$

Work-kinetic energy theorem for rotational motion: $W_{tot} = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2$

Power of a torque: $P = \tau\omega$

8 Angular momentum

Condition for rolling without slipping

- velocity of wheel's center of mas): $v_{cm} = R\omega$
- Linear acceleration of the center of mass: $a_{cm} = R\alpha$
- Distance travelled: $d_{cm} = R\theta$
- Acceleration of an object: $a_{cm} = \frac{mgsin\theta}{m+(I_{cm}/r^2)}$

Angular momentum for a particle: $\overrightarrow{L} = \overrightarrow{r} \times \overrightarrow{p} = \overrightarrow{r} \times m\overrightarrow{v}$

$L = rpsin\theta = rmvsin\theta$

Angular velocity of rigid body: $L = I\omega$

Precession angular speed for a gyroscope: $\Omega = \frac{F_g r}{I\omega}$, where F_g is the gravitational force acting on the gyroscope and r is the radius of precession.

(Gyroscope - a device used for measuring or maintaining orientation and angular velocity.)

9 Harmonic motion

9.1 Simple harmonic motion

Condition for simple harmonic frequency: $F_x = -kx$

angular frequency: $\omega = \sqrt{\frac{k}{m}}$

frequency: $f = \frac{1}{2\pi}\sqrt{km}$

period: $T = 2\pi\sqrt{\frac{m}{k}}$

where m is a mass of the oscillating system and k is force constant

$x(t) = Acos(\omega t + \phi)$, $x_{max} = A$

$v(t) = x'(t) = -A\omega sin(\omega t + \phi)$, $v_{max} = A\omega$

$a(t) = v'(t) = -A\omega w^2 cos(\omega t + \phi)$, $a_{max} = A\omega^2$

where A is amplitude and ϕ is phase angle

Energy: $E = U + K = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2$, in this case U acts as $cos^2\theta$ and K acts like $sin^2\theta$

$|v| = \pm\sqrt{\frac{k}{m}(A^2 - x^2)}$

9.2 Pendulums

Force equation for a simple pendulum: $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$

Angular frequency for a simple pendulum: $\omega = \sqrt{\frac{g}{L}}$

Period of a simple pendulum: $T = 2\pi\sqrt{\frac{L}{g}}$

$f = \frac{1}{2\pi}\sqrt{g}L$

where L is length of a string

Angular frequency for a torsional pendulum: $\omega = \sqrt{\frac{\kappa}{I}}$

frequency of a torsional pendulum: $f = \frac{1}{2\pi}\sqrt{\kappa}I$

period of a torsional pendulum: $T = 2\pi\sqrt{\frac{I}{\kappa}}$

where κ is the torsion constant and I is the moment of inertia

Angular frequency for a physical pendulum: $\omega = \sqrt{\frac{mgd}{I}}$

Frequency of a physical pendulum: $f = \frac{1}{2\pi}\sqrt{mgd}I$

Period of a physical pendulum: $T = 2\pi\sqrt{\frac{I}{mgd}}$

where d is the distance between the center of gravity and the axis of rotation

9.3 Oscillations

Newton's second law for harmonic motion: $m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$

Solution for underdamped harmonic motion: $x(t) = A_0 e^{-\frac{b}{2m}t} cos(\omega t + \phi)$, where b is damping constant

$\alpha = \frac{b}{2m}$

Natural angular frequency of a mass-spring system (driving frequency): $\omega_0 = \sqrt{\frac{k}{m}}$

Angular frequency of underdamped harmonic motion: $\omega = \sqrt{\omega_0^2 - (\frac{b}{2m})^2}$

Newton's second law for forced, damped oscillation: $-kx - b\frac{dx}{dt} + F_0 sin(\omega t) = m\frac{d^2x}{dt^2}$

Solution to Newton's second law for forced, damped oscillations: $x(t) = Acos(\omega t + \phi)$

Amplitude of system undergoing forced, damped oscillations: $A = \frac{F_0}{\sqrt{m^2(\omega^2 - \omega_0^2)^2 + b^2\omega^2}}$

10 Thermodynamics

Farenheit to Celcius: $T_c = \frac{5}{9}(T_F - 32°F)$

Celcius to Farenheit: $T_F = \frac{9}{5}T_C + 32°F)$

Thermal expansion: $\Delta L = \alpha \Delta T L_O$ where α is coefficient of linear expansion

Area expansion: $\Delta A = 2\alpha A \Delta T$

Volume expansion: $\Delta V = 3\alpha V \Delta T = \beta V \Delta T$

Heat: $Q = cm\Delta$, where c is specific heat capacity; $Q = nC\Delta T$, where C is molar heat capacity and n is number of moles

Heat for phase change: $Q = \pm mL$ where L is latent heat of fusion, vaporization or sublimation. If only part of the substance undergoes a phase change: $Q \pm xmL$ where x is the fraction that undergoes a phase change.

Heat flow between objects isolated from their surroundings: $\sum Q = 0$

Rate of heat transfer (heat current): $P = \frac{dQ}{dt} = \frac{kA(T_H - T_C)}{d}$ where **k** is thermal conductivity, **d** is distance between objects(the length of the heat flow path), **A** is the area through which the heat flows,$\mathbf{T}_\mathrm{H}$ is the temperature of the hot end, and $\mathbf{T}_\mathrm{C}$ is the temperature of cold end

Radiation of heat:

- net heat transfer rate: $P_{net} = \sigma e A(T^4 - T_s^4)$ where **T** is the temperature of the object and $\mathbf{T}_\mathrm{s}$ is the temperature of the surroundings
- Stephan-Boltzmann law: $P = \sigma A e T^4$
- where $\sigma = 5.67 \cdot 10^{-8} J/(s \cdot m^2 \cdot K^4)$ or $\sigma = 5.67 \cdot 10^{-8} W/(m^2 \cdot K^4)$ is called the Stefan-Boltzmann constant
- where e is the emissivity (object's effectiveness in emitting energy as thermal radiation) of the object

10.1 The kinetic theory of gasses

1atm$= 1.013 \cdot 10^5 Pa = 1.013 bar$

$Pa = \frac{N}{m^2}$

$\mathbf{k}_\mathrm{B} = 1.39 \cdot 10^{-23} J/K = R/N_A$

$\mathbf{R} = 8.31 J/(mol \cdot K) = 0.0821 \frac{L \cdot atm}{mol \cdot K}$

Pressure: $p = \frac{dF}{dA}$ where F is perpendicular to the surface

Ideal gas law: $pV = Nk_BT = nRT = nk_BN_AT$, where Boltzmann constant - $\mathbf{k}_\mathrm{B} = 1.39 \cdot 10^{-23} J/K$ and universal gas constant - $\mathbf{R} = 8.31 J/(mol \cdot K) = 0.0821 \frac{L \cdot atm}{mol \cdot K}$, **n** is number of moles, **N** is number of particles (atoms or molecules), and $\mathbf{N}_\mathrm{A} = 6.022 \cdot 10^{23} mol^{-1}$

Dalton's law: $P_{total} = \sum_{i=1}^{n} P_i$

Comparing two states of a constant mass of the ideal gas: $\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$

Microscopic equation of states: $pV = \frac{mN}{3}\overline{v^2}$ where $\overline{v^2}$ is the average squared speed

$\overline{v^2} = \frac{3k_BT}{m} \Rightarrow v_{RMS} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{8RT}{\pi M}}$

Average kinetic energy of a molecule: $\overline{K} = \frac{3}{2}k_BT$

Internal energy for ideal (monatomic gas): $E_{Total} = \frac{3}{2}Nk_BT = \frac{3}{2}\frac{m}{M}T$

Internal energy for diatomic gas: $E = \frac{5}{2}Nk_BT$

Equipartition theorem: there is always $\frac{k_BT}{2}$ per degree of freedom (degree of freedom shows how the particle can move, for example, along x,y,z axes, so df=3)

$p = \frac{1}{3}m_0n\overline{v^2}$

$p = \frac{mN}{LA} \cdot \frac{\overline{v^2}}{3}$

Maxwell-Boltzmann distribution of molecular speeds: $f(v) = \frac{4}{\sqrt{\pi}}(\frac{m}{2k_BT})^{3/2}v^2e^{-mv^2/(2k_BT)}$

Mean free path of molecules in an ideal gas: $\lambda = vt_{mean} = \frac{V}{4\pi\sqrt{2}r^2N}$, where r is the molecular radius

Mean free time: $\tau = \frac{k_BT}{4\sqrt{2}\pi r^2pv_{rms}}$

Van der Waals equation of states: $[p + a(\frac{n}{V})^2](V - nb) = nRT$

Heat capacity for monatomic gas: $C_V = \frac{3}{2}R$

Heat capacity for diatomic gas: $C_V = \frac{5}{2}R$

Heat capacity for monatomic solid: $C_V = 3R$

10.2 The first law of thermodynamics

The first law of thermodynamics: $\Delta U = Q - W$, where $W = \int_{V_1}^{V_2} PdV$, and W is work done by the system

when system does work on something we have W>0 and if work was done on the system when W<0

Internal energy change in an ideal gas: $\Delta E_{int} = nC_V\Delta T$

Ratio of heat capacity: $\gamma = \frac{C_p}{C_V}$, where C_P is the molar heat at constant pressure and C_V is the molar heat capacity at constant volume

isobaric: $\Delta p = 0$

- $E_{total} = Q - p\Delta V$
- in ideal gas: $\frac{V_A}{T_A} = \frac{V_B}{T_B}$
- infinitesimal transformation: $dQ = dE_{total} + pdV = nC_VdT + nRdT = n(C_V + R)dT$, where $dE_{total} + pdV = dH$ - enthalpy, and $C_P = C_V + R$ where C_P is the molar heat at constant pressure and C_V is the molar heat capacity at constant volume

isochoric: $\Delta V = 0$

- $W = 0$
- $Q = \Delta E_{total} = nC_V(T_B - T_A)$
- in ideal gas: $\frac{P_A}{T_A} = fracP_BT_B$

isothermal: $\Delta T = 0$

- $\Delta E_{total} = 0 \rightarrow Q = W$
- in ideal gas: $P_A V_A = P_B V_B$
- if reversible: $\Delta W_{AB} = \int_A^B PdV = nRTln(\frac{V_B}{V_A})$ and $\Delta W_{AB} = nRTln(\frac{p_A}{p_B})$
- expansion: W>0 and Q>0
- compression: W<0 and Q<0

adiabatic: $Q = 0$

- $\Delta E_{total} = -W = nC_V(T_B - T_A)$
- work for ideal gas: $W = nC_V(T_1 - T_2) = \frac{C_V}{R}(p_1V_1 - p_2V_2) = \frac{1}{\gamma-1}(p_1V_1 - p_2V_2)$
- if reversible: $dT = -\frac{pdV}{nC_V}$
- for ideal gas: $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$
  $p_1V_1^{\gamma} = p_2V_2^{\gamma}$
  $p_1^{1-\gamma}T_1^{\gamma} = p_2^{1-\gamma}T_2^{\gamma}$

10.3 The second law of thermodynamics

closed loop: $\Delta E_{total} = 0$ and $W = Q_H - Q_C$

efficiency of an engine:

- heat engine: $e = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H}$
- refrigerator: $K_R = \frac{Q_C}{W} = \frac{Q_C}{Q_H - Q_C}$
- heat pump: $K_P = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_C}$

efficiency in the Otto cycle: $e = 1 - \frac{1}{r^{\gamma-1}}$

Carnot engine:

- Heat transfer: $\frac{Q_C}{Q_H} = -\frac{T_C}{T_H}$
- efficiency: $e = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$
- coefficient of performance: $K = \frac{T_C}{T_H - T_C}$

Entropy change for a reversible process: $\Delta S = \int_1^2 \frac{dQ}{T}$ where 1 is initial and 2 is final states. If the process is isothermal: $\Delta S = \frac{Q}{T}$

Entropy change for an object undergoing a temperature change: $\Delta S = mcln(\frac{T_2}{T_1})$

Entropy in terms of microstates: $S = kln(w)$ where w is the number of microstates

Entropy change in terms of microstates: $\Delta S = kln(\frac{w_2}{w_1})$ Entropy of a system undergoing any complete reversible cyclic process: $\oint dS = \oint \frac{dQ}{T} = 0$

Change of entropy of a closed system under an irreversible process: $\Delta S \geq 0$

Change in entropy of the system along an isotherm: $lim_{T\to 0}(\Delta S)_T = 0$

11 Gravitation

Gravitational law: $F = G\frac{m_1 m_2}{r^2}$, where gravitational constant $G = 6.67 \cdot 10^{-11} Nm^2/kg^2$, r is the distance between the two objects, and m_1, m_2 are the masses of two objects

Multiple sources: $F_{total} = \sum_j F_{ij}$

$g = \frac{GM}{R^2}$ where M is the mass of the planet and R is the radius of the planet

Gravitational potential energy: $E_{pot,g} = -fracGm_1m_2r$

Speed in circular orbit: $v = \sqrt{\frac{GM}{r}}$

Period in circular orbit: $T = \frac{2\pi r}{v} = \frac{2\pi r^{3/2}}{\sqrt{GM}}$, where v is the speed of orbit, M is the mass of the planet, r is the radius of orbit

$(\frac{T_1}{T_2})^2 = (\frac{R_1}{R_2})^3$

escape velocity: $v = \sqrt{\frac{2GM}{R}}$

Conservation of energy: $\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$

Gravitational potential energy beyond Earth: $U = -\frac{GM_Em}{r}$

Energy in circular orbit: $E = K + U = -\frac{GmM_E}{2r}$

Conic sections: $\frac{\alpha}{r} = 1 + ecos\theta$, where e is eccentricity, r is the distance between the planet and the Sun, and θ is the angle measured from the x-axis, which is along the major axis of the ellipse

Kepler's third law: $T^2 = \frac{4\pi^2}{GM}a^3$

if we assume circular orbit: $T = 2\pi\sqrt{\frac{r^3}{GM_E}}$

Schwarzschild radius: $R_S = \frac{2GM}{c^2}$, where c is escape velocity

12 Electric charges and fields

$e = 1.602 \cdot 10^{-19}C$ C-Coulomb

Coulomb's law: $F_{12} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1q_2|}{r_{12}^2}$ where $\varepsilon_0 = 8.85 \cdot 10^{-12}C/Nm^2$

$F_{12} = k_e\frac{|q_1q_2|}{r_{12}^2}$ where $k_e = 8.99 \cdot 10^9 Nm^2/C^2$

Superposition of electric forces: $F = \frac{1}{4\pi\varepsilon_0}Q\sum_{i=1}^{N}\frac{q_i}{r_1^2}$

Electric force due to an electric field: $F = QE$

Electric field at point P: $F = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{q_i}{r_1^2}$

Linear charge: $E = \frac{1}{4\pi\varepsilon}\int_{line}\frac{\lambda dl}{r^2}\hat{r}$, where λ is linear charge density

Surface charge: $E = \frac{1}{4\pi\varepsilon}\int_{surface}\frac{\sigma dA}{r^2}\hat{r}$, where σ is surface charge density

Volume charge: $E = \frac{1}{4\pi\varepsilon}\int_{volume}\frac{\rho dV}{r^2}\hat{r}$, where ρ is volume charge density

Field of an infinite wire: $E = \frac{1}{4\pi\varepsilon_0}\frac{2\lambda}{z}$

Field of an infinite plane: $E = \frac{\sigma}{2\varepsilon_0}$

Electric potential energy: $-\Delta U = \Delta K = W$, where $W = \int_{p_1}^{p_2} \overrightarrow{F} \cdot d\overrightarrow{l}$

Work done to assemble a system of charge: $W_{1,2,...,N} = \frac{k}{2}\sum_j^N \sum_i^N \frac{q_i q_j}{r_{ij}}$,where $i \neq j$

13 Electric potential and current

$1eV = 1.6 \cdot 10^{-19} J$

Potential energy in two-charge system: $U = -k\frac{qQ}{r}$

Electric potential: $V = \int_R^P \overrightarrow{E} \cdot d\overrightarrow{l} = \frac{U}{q}$

Potential difference: $\Delta V = \frac{\Delta U}{q}$ or $\Delta U = q\Delta V$

$W = -q\Delta V$

Electric field: $\overrightarrow{E} = -\nabla V$ (gradient of V), $E = \frac{\Delta V}{d}$

Potential difference between two points: $\Delta V_{AB} = V_B - V_A = -\int_A^B \overrightarrow{E} \cdot d\overrightarrow{l}$

Electric potential of a point charge: $V = \frac{kq}{r}$

Electric potential of a system of point charges: $V_P = k\sum_1^N \frac{q_i}{r_i}$

Electric dipole moment: $\overrightarrow{p} \equiv q\overrightarrow{d}$

Electric potential for uniform charge disk: $V_P = \int_R^P \overrightarrow{E} \cdot d\overrightarrow{l} = \int dV_p = k2\pi\sigma \int_0^R \frac{rdr}{\sqrt{z^2+r^2}} = k2\pi\sigma(\sqrt{z^2+R^2} - \sqrt{z^2})$; $dq = \sigma 2\pi r dr$; $dV_p = k\frac{dq}{z^2+r^2}$

Electric potential due to a dipole: $V_P = k \cdot \frac{\overrightarrow{p}\cdot\hat{r}}{r^2}$

When r»d: $V_P = k \cdot \frac{\overrightarrow{p}\cdot\hat{r}}{r^2}$; $\frac{1}{r} \sim \frac{1}{\sqrt{1 \pm x}} \sim 1 \pm \frac{x}{2}$

At the surface of sphere: $V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{R}$

Electric power: $P = \frac{\Delta U}{\Delta t} = \frac{q\Delta V}{\Delta t} = IV = I^2 R = \frac{V^2}{R} = \frac{\varepsilon}{l} = \frac{W}{t} = \frac{Fd}{t} = \frac{qEd}{t}$

Definition of an ampere: $1A = 1C/s$

Electric current: $I = \frac{dQ}{dt} = qnAv_d$

Drift velocity: $v_d = \frac{I}{nqA}$

Current density: $I = \int\int_{area} \overrightarrow{J} \cdot d\overrightarrow{A}$; $\overrightarrow{J} = \sigma\overrightarrow{E}$

Resistivity: $\rho = \frac{1}{\sigma} = \frac{E}{J}$, where σ is conductivity

Common expression of Ohm's law: $V = IR$

Resistivity as a function of temperature: $\rho = \rho_0[1 + \alpha(T - T_0)]$

Definition of resistance: $R = \frac{V}{I}$

Resistance of a cylinder of material: $R = \rho\frac{L}{A}$

Temperature dependence of resistance: $R = R_0(1 + \alpha\Delta T)$

Torque on dipole in external E-field: $\overrightarrow{\tau} = \overrightarrow{p} \times \overrightarrow{E}$

14 Magnetic forces and fields

1 Gaus=10^{-4}Tesla

$[T] = \frac{[N]}{[A \cdot m]}$

$B_{earth} \sim 5 \cdot 10^{-5} Tesla$; $B_{strongmagnet} \sim 2Tesla$

Force on a charge in a magnetic field: $\overrightarrow{F} = q\overrightarrow{v} \times \overrightarrow{B}$

Magnitude of magnetic force: $F = qvBsin\theta$

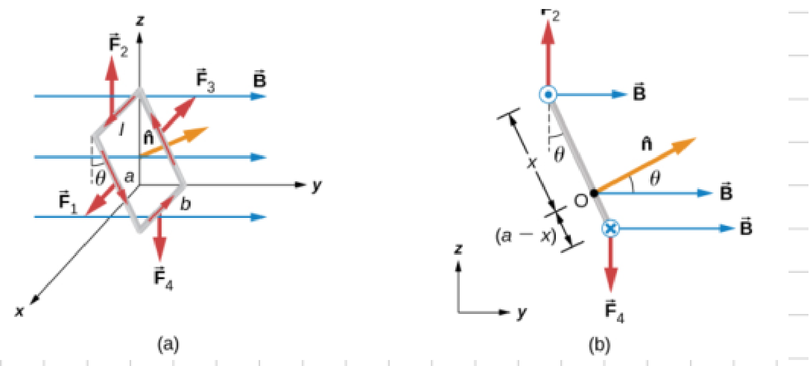
Radius of particle's path in a magnetic field: $R = \frac{mv}{qB}$

Period of a particle's motion in a magnetic field: $T = \frac{2\pi m}{qB} = \frac{2\pi R}{v}$

Force on a single charge: $d\overrightarrow{F} = nAev_d d\overrightarrow{l} \times \overrightarrow{B} \Rightarrow \overrightarrow{F} = I\overrightarrow{l} \times \overrightarrow{B}$

Force on a circular wire:

- if we are just considering a small segment: $dF = IBsin\theta dl = IBRsind\theta$
- force on the upper semicircle: $F_{upper} = 2IBR$
- force on the lower semicircle: $F_{lower} = -2IBR$



- $\overrightarrow{F}_1 = IABcos\theta\hat{i}$
- $\overrightarrow{F}_3 = -IABcos\theta\hat{i}$
- $\overrightarrow{F}_2 = IAB\hat{k}$
- $\overrightarrow{F}_3 = -IAB\hat{k}$

Torque: $\tau_{total} = -IabBsin\theta\hat{i}$, where ab is area of the loop

Magnetic moment/ magnetic dipole moment: $\overrightarrow{\mu} = IA\hat{n}$, where $\hat{n}$ is normal vector

$\overrightarrow{\tau} = \overrightarrow{\mu} \times \overrightarrow{B}$

$U = -\overrightarrow{\mu}\overrightarrow{B}$

15 Application of magnetic field and forces

Drift velocity in crossed electric and magnetic fields: $v_d = \frac{E}{B}$

Hall potential: $V = \frac{IBl}{neA} = Blv_d$

Charge-to-mass ration in a mass spectrometer: $\frac{q}{m} = \frac{E}{BB_0R}$ (the ratio can tell what the ions are)

Maximum speed of a particle in a cyclotron: $v_{max} = \frac{qBR}{m}$

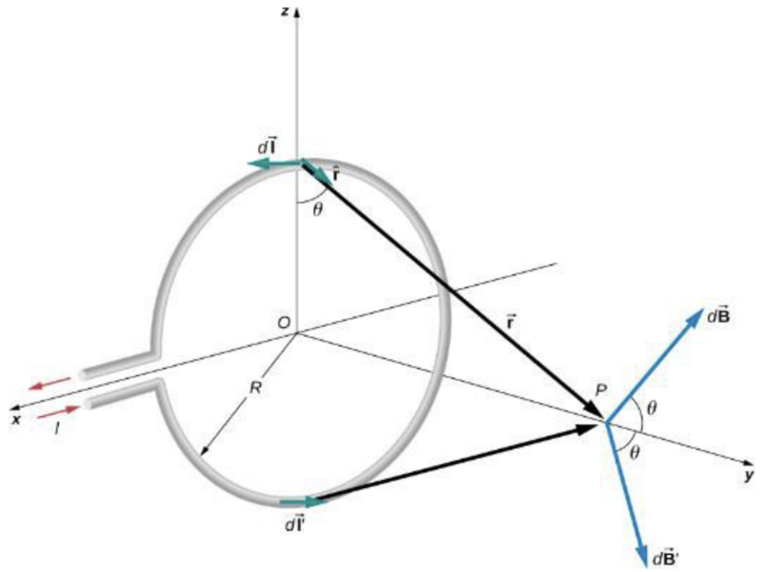
Biot-Savart law: $\overrightarrow{B} = \frac{\mu_0}{4\pi}\int_{wire} \frac{Id\overrightarrow{l}\times\hat{r}}{r^2}$, where $\mu_0 = 4\pi \cdot 10^{-7}Tm/A$

Contrbution to magnetic field from a current element: $dB = \frac{\mu_0}{4\pi}\frac{Idlsin\theta}{r^2}$

Magnetic field due to a long straight wire: $B = \frac{\mu_0 I}{2\pi R}$

Magnetic force between two parallel currents: $F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$

16 Sources of magnetic field



Magnetic field of a current loop (top): $dB = \frac{\mu_0}{4\pi}\frac{Idlsin(\pi/2)}{r^2} = \frac{\mu_0}{4\pi}\frac{Idl}{y^2+R^2}$

$B = \frac{\mu_0}{2}\frac{IR^2}{(y^2+R^2)^{3/2}}j = \frac{\mu_0\mu\hat{j}}{2\pi(y^2+R^2)^{3/2}}$

Magnetic field of a current loop (at the center of loop): $B = \frac{\mu_0 I}{2R}$

flat coil n turns: $B = \frac{\mu_0 nI}{2R}$

y»R: $B = \frac{\mu_0\mu}{2\pi y^3}$

Ampere's law: $\oint \overrightarrow{B} \cdot d\overrightarrow{l} = \mu_0 I$

Magnetix field strength inside a solenoid: $B\frac{\mu_0 IN}{L} = \mu_0 nI$

Magnetic field strength inside a toroid: $B = \frac{\mu_0 NI}{2\pi r}$

Magnetic permeability: $\mu = (1+\chi)\mu_0$

Magnetic field of a solenoid filled with paramagnetic material: $B = \mu nI$

17 Electromagnetic induction

The emf (potential difference): $\varepsilon = \frac{dW}{dq}[J/C] = [V]$

$I = \frac{\varepsilon}{r+R}$

Motionally induced emf: $\varepsilon = Blv$

Magnetic flux: $\Phi_m = \int_S \overrightarrow{B} \cdot \hat{h} dA$

Faraday's law: $\varepsilon = -N \frac{d\Phi_m}{dt}$

Motional emf around a circuit: $\varepsilon = \oint \overrightarrow{E} \cdot d\overrightarrow{l} = \frac{d\Phi_m}{dt}$

Induced E-field outside the coil (r>R): $E = \frac{\alpha \mu_0 n I_0 R^2}{2r} e^{-\alpha t}$

Induced E-field inside the coil (r<R): $E = \frac{\alpha \mu_0 n I_0 r}{2} e^{-\alpha t}$

Emf produced by an electric generator: $\varepsilon = NBA\omega sin(\omega t)$, where $NBA\omega = \varepsilon_0$

$P = \frac{I^2 B^2 v^2}{R}$